

ME 445/545
Homework 3

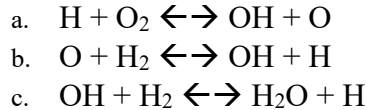
Submission guidelines: Must upload to the 445 or 545 gradescope pages. Must select corresponding pages, if not selected 5 points will be reduced.

In developing the skills to be able to “design a combustor” we need to be able to determine the time-scale and concentration of major and minor species within the combustion process. Because some species are “good” and others “bad (e.g., NO), we want to be able to predict which are present at the exit of our combustor. We also want to better understand the “reactivity” of our mixture. The problems listed below will introduce us to solving for chemical kinetic related problems.

1. (Turns 4.8) Classify the following reactions as either global or elemental. For those identified as elementary, further classify them as unimolecular, bimolecular, or termolecular.
 - a. $\text{CO} + \text{OH} \rightarrow \text{CO}_2 + \text{H}$
 - b. $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
 - c. $\text{H}_2 + \text{O}_2 \rightarrow \text{H} + \text{H} + \text{O}_2$
 - d. $\text{HOCO} \rightarrow \text{H} + \text{CO}_2$
 - e. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 - f. $\text{OH} + \text{H} + \text{M} \rightarrow \text{H}_2\text{O} + \text{M}$
2. (Turns 4.4) Consider the reaction $\text{CH}_4 + \text{O}_2 \rightarrow \text{CH}_3 + \text{HO}_2$. Although a CH_4 molecule may collide with an O_2 molecule, a reaction may not necessarily occur. List two important factors in determining whether or not a reaction occurs during collision.
3. (Turns 4.10) In methane combustion, the following reaction pair is important:
$$\text{CH}_4 + \text{M} \rightleftharpoons \text{CH}_3 + \text{H} + \text{M}$$
Where the reverse reaction coefficient is given by:
$$k_r \text{ (m}^6\text{/kmol}^2\text{s)} = 2.82 \times 10^5 T \exp[-9835/T]$$
At 1500K, the equilibrium constant K_P has the value of 0.003691 based on a reference-state pressure of 1 atm (101,325 Pa). Derive an algebraic expression for the forward rate coefficient k_f . Evaluate your expression for a temperature of 1500K. **Include units.**
4. (Turns 4.14) Consider the following elementary reaction mechanism where both the forward and reverse reactions are important:
 1. $\text{CO} + \text{O}_2 \rightleftharpoons \text{CO}_2 + \text{O}$
 2. $\text{O} + \text{H}_2\text{O} \rightleftharpoons \text{OH} + \text{OH}$
 3. $\text{CO} + \text{OH} \rightleftharpoons \text{CO}_2 + \text{H}$
 4. $\text{H} + \text{O}_2 \rightleftharpoons \text{OH} + \text{O}$

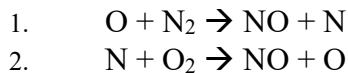
How many chemical rate equations are needed to determine the chemical evolution of a system defined by this mechanism? Write out the rate equation for the hydroxyl radical.

5. (Turns 4.20) In the combustion of hydrogen, the following reactions involving radicals are fast in both the forward and reverse directions:



Use the assumption of partial equilibrium to derive algebraic expressions for the molar concentrations of the three radical species O, H, and OH, in terms of the kinetic rate coefficients and the molar concentrations of reactant and product species, H_2 , O_2 , and H_2O .

6. GRAD: (Turns 4.16) Consider the following chain-reaction mechanism for the high temperature formation of nitric-oxide:



- A. Write out expressions for $d[\text{NO}]/dt$ and $d[\text{N}]/dt$
- B. Assuming N atoms exist in steady state and that the concentrations of O, O_2 , and N_2 are at their equilibrium values for a specified temperature and composition, simplify your expression obtained above for $d[\text{NO}]/dt$ for the case of negligible reverse reactions.
- C. Write out the expression for the steady state N atom concentration used in part B.
- D. For the conditions given below and using the assumptions of part B, how long does it take to form 50 ppm (mole fraction $\times 10^6$) of NO?

$$T = 2100 \text{ K}; \rho = 0.167 \text{ kg/m}^3; \text{MW} = 28.778; \chi_{\text{O}} = 7.6 \times 10^{-5};$$

$$\chi_{\text{O}_2} = 3.025 \times 10^{-3}; \chi_{\text{N}_2} = 0.726;$$

$$k_{1f} = 1.82 \times 10^{14} \exp(-38,370/T) \text{ [units of cm}^3/\text{gmol s]}$$

$$k_{2f} = 1.8 \times 10^{10} T \exp(-4680/T) \text{ [units of cm}^3/\text{gmol s]}$$

- E. Calculate the value of the reverse reaction rate coefficient for the first reaction for a temperature of 2100K
- F. For your computations in Part D, how good is the assumption that the reverse reaction (for reaction 1 only) is negligible? Be quantitative. Hint compare N production and destruction for reaction 1.
- G. For the conditions of Part D, determine numerical values for $[\text{N}]$ and χ_{N} .

HW 3 Solution ranges

- 1) a)
- b)
- c)
- d)
- e)
- f)

- 2) Reason 1
- Reason 2

- 3) 17 - 19 $\text{m}^3/\text{kmol}\cdot\text{s}$

- 4) a)
- b)

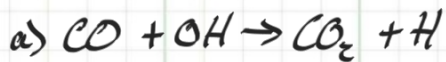
- 5) a: $[\text{O}] =$
- b: $[\text{H}] =$
- c: $[\text{OH}] =$

- 6) Grad A) $d[\text{NO}]/dt =$
- $d[\text{N}]/dt =$
- B) $d[\text{NO}]/dt =$
- C) $[\text{N}] =$
- D) $t = 0.03 - 0.04 \text{ s}$
- E) $k = 3. - 3.3 \times 10^{13} \text{ (cm}^3/\text{gmol}\cdot\text{s)}$
- F) forward/ reverse 7 - 8
- G) X_{N} 9 - $9.8 \times 10^{-3} \text{ ppm}$
- $[\text{N}]$ 5 - $5.5 \times 10^{-11} \text{ kmol/m}^3$

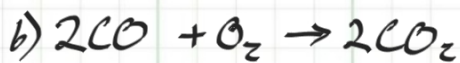
Problem #1

- Classify the following Rx as Global or Elementary

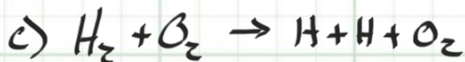
* if Rx is Elementary classify as unimolecular, bimolecular or termolecular



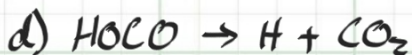
This is an Elementary Bimolecular Rx



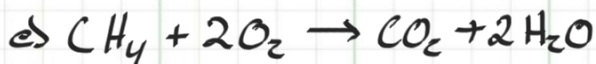
This is a global Rx because the Rx between it would more than likely take multiple interactive bimolecular steps to split O_2 into a free radicals then combine to form 2CO_2 . This is more likely to happen than a single step elementary termolecular Rx.



This is an Elementary Bimolecular Rx where the collisions between H_2 & O_2 decompose H_2 into $\text{H} + \text{H}$ free radicals



This is an Elementary unimolecular Rx where HOCO decomposes into $\text{H} + \text{CO}_2$



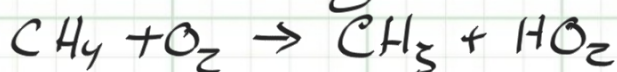
This a global Rx $\because$ Fuel and Oxidizer forms Combustion Products.



This is an Elementary Termolecular Rx

Problem #2

Consider The following Rx -



Although CH_4 may collide with O_2 , the molecular Rx may not occur.

List 2 important factors in determining if a Rx occurs during collision.

* The Two important factors that determine if a Rx occurs during a collision is:

* The Activation Energy or the minimum Energy required to incite a molecular Rx to take place

* The Steric Factor p - This factor accounts for geometrical interactions of the collision.

(However These factors can not be determined by Collision Theory)

Therefore we take the pre-exponential factor (i.e. the Frequency factor) of collision and Activation Energy from tabulations done experimentally based on Temperature.

Standardized form $k(T) = AT^b e^{\left(\frac{-E_a}{RT}\right)}$ ^{Empirically found}

Problem #3

- For methane combustion -

The Rx pair $\text{CH}_4 + \text{M} \rightleftharpoons \text{CH}_3 + \text{H} + \text{M}$ is important

The Reverse Rx Coef. is given by -

$$K_r (\text{m}^3/\text{kmol}^2\text{s}) = 2.82 \times 10^5 T e^{\left(\frac{-9835}{T}\right)}$$

@ $T = 1500\text{K}$, the equilibrium $K_p = 0.003691$ based on $P_{\text{ref}} = 1 \text{ atm}$

* Derive an algebraic expression for the forward rate coef. K_f

* Evaluate K_f @ $T = 1500\text{K}$ (include units)

Considering, $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D} + \text{M}$ and reaction is balanced

$$K_p = K_c (R_u T / P^\circ)^{c+d+m-a-b}$$

$$\Rightarrow K_p = K_c (R_u T / P^\circ)^1 \Rightarrow K_c = \frac{K_p P^\circ}{R_u T}$$

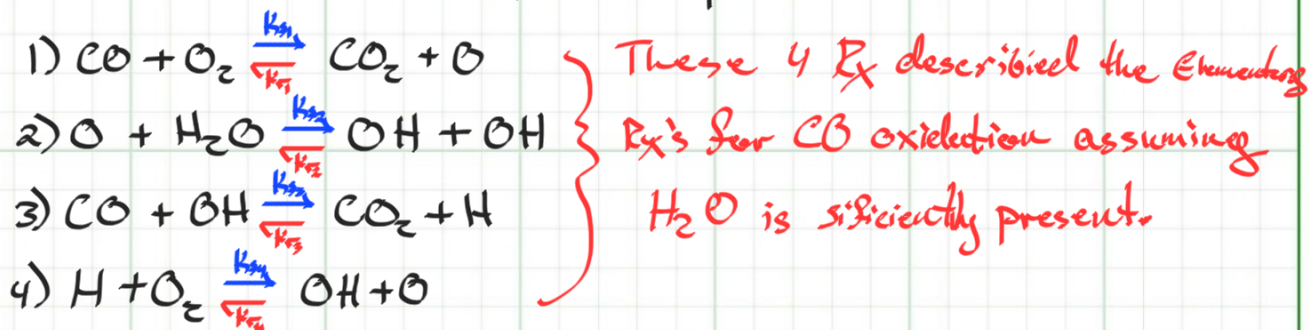
$$\frac{K_f}{K_r} = K_c \Rightarrow K_f = \frac{K_r K_p P^\circ}{R_u T}$$

$$K_f = \frac{(2.82 \times 10^5 (1500) e^{\left(\frac{-9835}{1500}\right)}) \text{ m}^3/\text{kmol}^2\text{s} (0.003691) (101.325 \text{ kPa})}{(8.314 \text{ kJ/kmol}^\circ\text{K}) (1500^\circ\text{K})}$$

$$K_f = 18.021 \text{ m}^3/\text{kmol}\cdot\text{s}$$

Problem # 4

Consider the following Elementary Mechanisms were both forward and reverse k_x are important.



* Determine How many chemical rate equations are required to determine chemical evolution of a system defined by the mechanism's
 - Write out the rate equation for the hydroxy radical.

To determine the quantity of chemical rate equations we have to determine the # of reactive species in the Mechanism / rate equation for each reactive species.

- There are 7 reactive species due to forward & reverse Rx's being important.

$$\frac{d[\text{CO}]}{dt}, \frac{d[\text{CO}_2]}{dt}, \frac{d[\text{O}]}{dt}, \frac{d[\text{O}_2]}{dt}, \frac{d[\text{H}_2\text{O}]}{dt}, \frac{d[\text{OH}]}{dt}, \frac{d[\text{H}]}{dt}$$

Note concentration of species.

$$\frac{d[X_i]}{dt} = \sum (\text{Rate of production of OH}) - \sum (\text{Rate of destruction of OH})$$

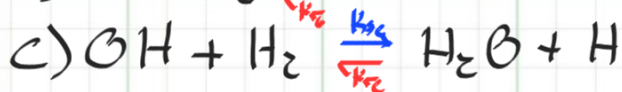
$$\frac{d[\text{OH}]}{dt} = (2k_{f2}[\text{H}_2\text{O}][\text{O}] + k_{r3}[\text{CO}_2][\text{H}] + k_{f4}[\text{H}][\text{O}_2])$$

$$- (2k_{r2}[\text{OH}]^2 + k_{f3}[\text{CO}][\text{OH}] + k_{r4}[\text{OH}][\text{O}])$$

2 implies 2 species produced / destroyed
 * Note $\rightarrow [\text{OH}][\text{OH}] = [\text{OH}]^2$

Problem #5

- For the Combustion of hydrogen, The following Rx's involving radicals are fast in both forward and reverse directions



Bimolecular Rx Pairs

Assume: - Partial equilibrium

* Derive:

- Algebraic expressions for the Molar Concentration of the 3 radical species

- O, H, OH

in terms of the kinetic rate coef. and Molar Concentrations of reactants and product species

- H_2, O_2, H_2O

Considering fast radicals and Partial Equilibrium implies S.S.
i.e. $\frac{d}{dt} \approx 0$

$$\text{For, } K_{p3a} [H][O_2] = K_{pr3a} [OH][O]$$

$$K_{p3b} [O][H_2] = K_{pr3b} [OH][H]$$

$$K_{p3c} [OH][H_2] = K_{pr3c} [H_2O][H]$$

$$\Rightarrow \frac{K_{p3a}}{K_{pr3a}} = \frac{[OH][O]}{[H][O_2]} = K_{p3} \quad \frac{K_{p3b}}{K_{pr3b}} = \frac{[OH][H]}{[O][H_2]} = K_{p2}$$

$$\frac{K_{p3c}}{K_{pr3c}} = \frac{[H_2O][H]}{[OH][H_2]} = K_{p3}$$

Isolating Radicals

$$\text{for } [\text{O}] \quad [\text{O}] = \frac{k_{p1} [\text{H}] [\text{O}_2]}{[\text{OH}]} \quad (1)$$

$$\text{for } [\text{H}] \quad [\text{H}] = \frac{k_{p2} [\text{O}] [\text{H}_2]}{[\text{OH}]}$$

Plug [H] into (1)

$$[\text{O}] = \frac{k_{p1} k_{p2} [\text{O}] [\text{H}_2] [\text{O}_2]}{[\text{OH}]^2} \Rightarrow [\text{OH}]^2 = k_{p1} k_{p2} [\text{H}_2] [\text{O}_2] \\ \Rightarrow [\text{OH}] = \sqrt{k_{p1} k_{p2} [\text{H}_2] [\text{O}_2]}$$

$$[\text{H}] = \frac{k_{p2} [\text{OH}] [\text{H}_2]}{[\text{H}_2\text{O}]}$$

$$\Rightarrow [\text{H}] = \frac{k_{p2} \sqrt{k_{p1} k_{p2} [\text{H}_2] [\text{O}_2]} [\text{H}_2]}{[\text{H}_2\text{O}]} = \frac{k_{p2} \sqrt{k_{p1} k_{p2} [\text{O}_2]} [\text{H}_2]^{3/2}}{[\text{H}_2\text{O}]}$$

$$\Rightarrow [\text{O}] = \frac{k_{p1} k_{p2} \sqrt{k_{p1} k_{p2} [\text{O}_2]} [\text{H}_2]^{3/2} [\text{O}_2]}{\sqrt{k_{p1} k_{p2} [\text{H}_2] [\text{O}_2]}}$$

$$[\text{O}] = k_{p1} k_{p2} [\text{H}_2] [\text{O}_2]$$